

ADVANCING FOREX PRDEICTION WITH TRANSFORMER-BASED DEEP LEARNING MODELS

A PROJECT REPORT

20ISP01-PROJECT WORK II

Submitted by

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*In partial fulfilment of the requirements
for the award of the degree
of*

**MASTER OF SCIENCE
IN
SOFTWARE SYSTEMS**

DEPARTMENT OF COMPUTER TECHNOLOGY – PG



KONGU ENGINEERING COLLEGE

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APRIL 2025

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**ADVANCING FOREX PRDEICTION WITH TRANSFORMER-BASED DEEP LEARNING MODELS**” is the bonafide record of project work II (20ISP01) done by **GOKUL S (Register No: 20ISR014)** in partial fulfillment of the requirements for the award of the Degree of Master of Science in **SOFTWARE SYSTEMS** of Anna University, Chennai during the year 2024-2025.

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Submitted for the end-semester viva voce examination held on.....

INTERNAL EXAMINER

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DECLARATION

I affirm that the project Report titled “**ADVANCING FOREX PRDEICTION WITH TRANSFORMER-BASED DEEP LEARNING MODELS**” being submitted in partial fulfillment of the requirements for the award of Master of Science is the original work carried out by me. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on earlier occasion on this or any other candidate.

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I certify that the declaration made by the above candidate is true to the best of my knowledge.

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ABSTRACT

Foreign exchange rate forecasting is a critical component of the global financial market, influencing trading strategies, risk management, and economic policies. Given the highly volatile and complex nature of Forex markets, accurate forecasting remains a challenging task. Traditional time-series models often struggle to capture intricate temporal dependencies, market anomalies, and multi-factor influences, leading to suboptimal predictions. To enhance forecasting accuracy and decision-making, this study explores the application of deep learning models, specifically Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT).

These models offer distinct advantages: GRU provides an efficient recurrent structure for sequence modeling, Informer incorporates self-attention mechanisms to focus on long-term dependencies, and TFT integrates attention-based mechanisms with probabilistic forecasting for improved robustness. The dataset utilized in this research spans from 1970 to 2020, consisting of historical exchange rates and 86 additional financial indicators, ensuring a comprehensive representation of market trends. To prepare the data for optimal model performance, preprocessing steps include handling missing values through interpolation, removing extreme outliers, and normalizing features using Min-Max scaling.

This study aims to establish a comparative framework to assess the effectiveness of GRU, Informer, and TFT in capturing complex financial patterns. By leveraging deep learning techniques, this research seeks to improve short-term Forex forecasting and provide valuable insights into dynamic currency movements. The findings of this study will contribute to developing more reliable and sophisticated predictive models for financial analysts, traders, and policymakers, ultimately enhancing decision-making in the foreign exchange market.

ACKNOWLEDGEMENT

I express my sincere thanks to **Thiru A. K. Ilango B.Com., M.B.A., LLB.**, our Correspondent and all philanthropic trust members of Kongu Vellalar Institute of Technology Trust who has always encouraged me in the academic and co-curricular activities.

I am proudly thankful with no words of formal nature to our dynamic Principal **Prof. V. Balusamy B.E(Hons), M.Tech., PhD.**, for providing all the facilities indispensable during my project.

I am thankful to **Dr. E. Gothai MCA., MPhil., ME., PhD.**, Associate Professor and Head, Department of Computer Technology-PG, for her valuable suggestions and cooperation during my project work.

I am grateful to our project co-ordinator **Dr. P. Ananthi M.Sc., M.Phil., MCA., MBA., Ph.D.**, Associate Professor, Department of Computer Technology-PG, whose insight comments and suggestions added greatly to the final result.

I am grateful to my project guide **Mrs. K. Karthika B.E, M.E**, Assistant Professor, Department of Computer Technology-PG, for her valuable suggestions and cooperation during my project work.

TABLE OF CONTENTS

CHAPTERNO	TITLE	PAGE NO
	ABSTRACT	iv
	LIST OF TABLES	vii
	LIST OF FIGURES	viii
	LIST OF ABBREVIATIONS	ix
1	INTRODUCTION	1
	1.1 FOREX PREDICTION	1
2	LITERATURE SURVEY	3
3	SYSTEM REQUIREMENTS	5
	3.1 HARDWARE REQUIREMENTS	5
	3.2 SOFTWARE REQUIREMENTS	5
	3.3 SOFTWARE DESCRIPTION	5
4	PROPOSED METHODOLOGY	8
	4.1 INTRODUCTION	8
	4.2 MODULE DESCRIPTION	9
	4.3 ARCHITECTURAL DESIGN	16
5	RESULTS AND DISCUSSION	18
6	CONCLUSION AND FUTURE WORK	24
	APPENDICES	25
	APPENDIX 1:SAMPLE CODE	25
	APPENDIX 2: SAMPLE SCREENSHOTS	28
	REFERENCES	30

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
5.1	Performance Analysis	19

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
4.1	GRU Architecture	11
4.2	Informer Architecture	12
4.3	TFT Architecture	13
4.4	System Flow Architecture	17
5.1	Training and Validation Loss for GRU, Informer, and TFT	20
5.2	Actual vs. Predicted Exchange rates for GRU, Informer, and TFT	21
5.3	Actual vs. Predicted Percentage Changes (Clipped to $\pm 100\%$)	22
5.4	Evaluation Metrics for GRU, Informer, and TFT (MSE, MAE, Accuracy, Profitability)	23
A2.1	Dataset Before Preprocessing	28
A2.2	Dataset After Preprocessing	28
A2.3	Training and Validation Loss across Models	29
A2.4	Actual vs Predicted Exchange Rates (Test Set)	29

LIST OF ABBREVIATIONS

Forex	: Foreign Exchange
GRU	: Gated Recurrent Unit
TFT	: Temporal Fusion Transformer
MSE	: Mean Squared Error
MAE	: Mean Absolute Error
WMAPE	: Weighted Mean Absolute Percentage Error
INR	: Indian Rupee
USD	: United States Dollar
Min-Max	: Minimum-Maximum (Normalization)